A Comparison of Continuous Epidural, Spinal Opioid, and Patient-Controlled Analgesia for Posterior Spinal Fusion for Adolescent Idiopathic Scoliosis: A Multicenter, International Database Study

Lloyd M. Halpern, MD, De-An Zhang, MD,† Clark J. Kogan, PhD,‡ Kirk A. Ryan, CTS,§ Zackary Correll, BA,‖ and William Bronson, MD¶*

Background: There is conflicting evidence for the optimal method of analgesia following posterior spinal fusion for adolescent idiopathic scoliosis. The primary objective of this study is to test which method of postoperative analgesia (continuous epidural, spinal opioids, or patient-controlled analgesia) is associated with reduced opioid requirements, pain scores, complications, and length of stay.

Methods: This is a retrospective, multicenter study of the Shriners Children's International Pediatric Spine Database. Patients ages 13 to 19 years who had posterior spinal fusion with segmental spinal instrumentation for adolescent idiopathic scoliosis between January 01, 2011, and November 5, 2021, were eligible. Patients were divided into 3 cohorts based on the primary method of postoperative analgesia: continuous epidural (EPI), spinal opioid (SPI), or patient-controlled analgesia (PCA). We compared total parenteral and oral opioid usage, verbal numeric pain scores, complications, and the length of stay.

Results: A total of 2371 patients from 13 hospitals were included in the study. Total parenteral and oral opioid usage in the spinal and epidural groups was significantly lower compared with the patient-controlled analgesia group [OME/kg: SPI 1.7 (95% CI: 1.1-2.5) vs. EPI 1.9 (95% CI: 1.1-3.2) vs. PCA 4.1 (95% CI: 3.4-4.8): $P < 0.002$]. There were no clinically significant differences in mean daily pain scores, complications, or length of stay.

Conclusions: Spinal opioid and continuous epidural analgesia decrease parenteral and oral opioid requirements compared with patient-controlled analgesia. The increased opioid usage in the patient-controlled analgesia group does not lead to clinically significant differences in pain scores, complications or length of stay. We conclude that all 3 methods of analgesia provide safe and effective pain relief. This study reinforces the need for collaboration between pediatric orthopaedic surgeons and anesthesiologists to effectively manage postoperative pain following posterior spinal fusion for adolescent idiopathic scoliosis.

Level of Evidence: Level III—therapeutic study.

Key Words: scoliosis, spinal fusion, pain management, intrathecal injection, epidural analgesia, patient-controlled analgesia, postoperative recovery.

(*J Pediatr Orthop* 2026;00:000–000)

Adolescent idiopathic scoliosis is the most common spinal disorder in children. Posterior spinal fusion with segmental spinal instrumentation for scoliosis correction is associated with substantial pain and opioid use in the postoperative period. Continuous epidural analgesia and the administration of spinal opioids have been used to reduce opioid requirements and provide improved analgesia compared with patient-controlled analgesia. The results have been variable. Several studies have suggested that epidural analgesia and spinal opioids reduce pain scores following posterior spinal fusion,[1–3] while 3 studies and a meta-analysis found no differences in opioid usage or pain scores when comparing epidural analgesia with patient-controlled analgesia and spinal opioids.[4–7] The ideal analgesic method, or combination of methods, has not been identified. Previous studies comparing different analgesic modalities were performed at a single study location with relatively small sample sizes. The primary objective of this study is to use a large sample size from 13 different children's hospitals over 11 years to test the hypothesis that one of 3 methods of postoperative analgesia following posterior spinal fusion for adolescent idiopathic scoliosis is associated with decreased parenteral and oral opioid requirements, and that this leads to reduced pain scores, complications and length of stay. The

From the *Department of Anesthesia, Shriners Children's Hospital; ‡StatsCraft LLC; §Shriners Children's Hospital; ¶Department of Pediatric Orthopedic Surgery, Shriners Children's, Spokane; ‖University of Washington College of Medicine, Seattle, WA; and †Department of Anesthesia, Shriners Children's Hospital, Pasadena, CA.

Presented at the Society of Pediatric Anesthesia Annual Meeting, April 13, 2024, Anaheim, CA.

Support was provided solely from institutional and/or departmental sources.

The authors declare no conflicts of interest.

Reprints: Lloyd M. Halpern, MD, Department of Anesthesia, Shriners Children's Hospital, Spokane, WA. E-mail: lhalpern@shrinenet.org.

DOI: 10.1097/BPO.0000000000003272

primary outcome is opioid usage in the 3 treatment groups.

METHODS

This is a retrospective, multicenter study of the Shriners Children's International Pediatric Spine Database. Patients ages 13 to 19 years who had posterior spinal fusion with segmental spinal instrumentation for juvenile or adolescent idiopathic scoliosis at one of 13 Shriners Children's Hospital locations in the United States between January 01, 2011, and November 5, 2021, were included. Patients who were admitted more than 1 day before surgery, had previous spine surgery, required a reoperation during the same admission, had growing rods, anterior procedures or vertebrectomies were excluded. Approval was obtained for all institutions from the WCG Institutional Review Board. Patients were divided into 3 cohorts based on the primary method of postoperative analgesia: spinal opioid (SPI), continuous epidural (EPI), or patient-controlled analgesia (PCA). We collected demographic data for all patients to include age, sex, race, body mass index, hospital location, and preoperative diagnosis (idiopathic vs. nonidiopathic scoliosis). The number of vertebral levels fused was available in only 133 patients in the database. As a substitute, we determined the number of pieces of implanted durable hardware (ie, screws, rods). We found a good correlation with the number of implanted hardware and the levels fused (Fig. 1). Patients within 2 SDs from the median number of implanted hardware were included in the study (15 to 60 pieces).

Postoperative parenteral and oral opioid consumption was converted into oral morphine equivalent doses using standardized opioid conversion tables.[8] We compared total opioid usage (OME/kg) and the rate of opioid usage per hour (OME/kg/h) in the 3 treatment groups from departure from the operating room to hospital discharge, or up to 88 hours if not yet discharged, when 95% of the epidurals had been discontinued. Spinal and epidural opioids were not included in postoperative opioid totals. We queried the database for all medications administered postoperatively, including multimodal analgesia (acetaminophen, ibuprofen, and ketorolac).

We compared hourly mean verbal numeric pain scores (0 to 10) in the 3 treatment groups. Pain scores recorded as zero during sleep were not included. The length of stay was calculated from the time and date of admission to the time and date of hospital discharge. We determined the time to first documentation of ambulation and removal of intravenous and Foley catheters as a measure of enhanced recovery after surgery protocols. We documented hospital readmissions within 30 days of the surgical procedure.

We compared the hourly time spent with a complication in the 3 treatment groups. Postoperative complications were defined based on an International Classification of Diseases (ICD-9) diagnosis not present at admission or inferred by multiple dosing of counter-acting medications. The incidence of the following complications was determined: muscle spasm, nausea and vomiting, constipation, pruritus, and excessive sedation/respiratory depression. We a priori defined a clinically significant reduction in a complication rate as a statistically significant reduction in the entire 88-hour postoperative window and not just on a single postoperative day.

All epidural catheters were placed intraoperatively by the surgeon with the tip of the catheter measured to reach the fifth thoracic vertebral level. The composition and rate of the epidural infusion were highly variable. The epidural was discontinued ~60 hours after the out-of-operating-room time at 0500 on postoperative day 3. All spinal opioids were administered before incision by the anesthesiologist.

We powered our study, before the study onset, by considering the study design required to compare daily percentage opioid usage where treatments are applied at a hospital-level, hospital variability dominates, and hospital-level effects are normally distributed on the logit scale. For the calculation, we assumed usage levels of 98% in the opioid group and 50% in the epidural group with a facility-level SD of 1 on the logit (corresponding to 0.11 on the scale of the outcome around the assumed opioid group proportion).[9] We determined that a design that includes 13 facilities in the opioid group and 2 facilities in the epidural group is sufficient to obtain a power of 0.80 in comparing the proportion usage in the 2 groups.

Statistics

Hourly opioid usage, pain scores, number of hours reporting complications, time to ambulation, and length of stay were analyzed using linear mixed effects models. Opioid usage was truncated at 0.001 mg/kg/hour for daily averages and <0.024 mg/kg for end-of-day accumulated totals. Log transformations were applied to opioid usage and length of stay before analysis. All models were adjusted for treatment group, sex, age, body mass index, along with random effects for facility by day, facility by year and subject. Models with daily outcomes additionally included an interaction between the number of days postsurgery and the treatment group and a random effect for facility by day by group. The only exception was that due to fitting issues, the daily complications models were reduced to include only a facility by group interaction, rather than the facility by day by group interaction. Models were used to assess predictor significance, extract least squares means, and compare least squares means between groups within a day. A Tukey multiple comparisons adjustment was used for least squares group comparisons between groups within a day. All statistical analyses were performed using the R statistical software.[9] Linear mixed models were fit with the "lme4" package and the "emeans" package was utilized for estimating marginal means and conducting post hoc comparisons.[10]

RESULTS

A total of $N=9740$ pediatric patients undergoing

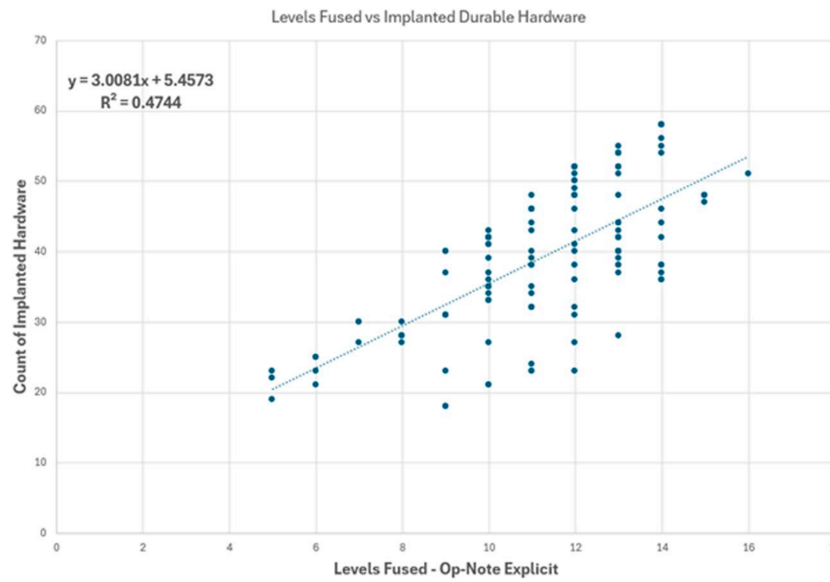

FIGURE 1. Linear correlation of the number of vertebral levels fused and the number of implanted durable hardware in 133 patients. *$P<0.05$. full color online

scoliosis surgery in 18 different International Shriners Children's Hospitals during the study period were identified.

Four hospitals in the patient-controlled analgesia group were removed because of too many errors in the recording of patient-controlled analgesia doses. One hospital in the epidural group had to be removed because of inconsistencies in the number of epidurals placed and systemic charting errors. The remaining 2371 pediatric patients from 13 pediatric orthopaedic hospitals satisfied all inclusion and exclusion criteria and were included in the study. One site had patients in both the spinal and opioid treatment groups. There were 263 children in the epidural group from one children's hospital, 2018 children in the patient-controlled analgesia group from 11 children's hospitals, and 90 children in the spinal group from 2 children's hospital locations. The demographic and covariate distribution across the 3 treatment groups are reported in Table 1.

For the primary outcome of parenteral and oral opioid usage, we found variability in hourly opioid usage at the hospitals in the patient-controlled analgesia group. We applied multilevel regression analyses to adjust for hospital-level variability to obtain reliable direct comparisons of mean opioid usage per day between treatment groups (Fig. 2A). Mean daily opioid usage in the spinal and epidural groups is strongly reduced compared with the patient-controlled analgesia group on the day of surgery, postoperative days 1 and 2. This continues postoperative day 3 in the spinal group. There is no statistically significant difference in mean daily opioid usage in the epidural and patient-controlled analgesia groups after epidural removal on postoperative day 3. There are no significant differences in opioid usage between the spinal and epidural groups on any day. Mean total opioid consumption for the 88-hour window is more than 2 times greater in the patient-controlled analgesia group [OME/kg; SPI: 1.7; 95% CI: 1.1-2.5 vs. EPI: 1.9; 95% CI: 1.1-3.2 vs. PCA: 4.1; 95% CI: 3.4-4.8; $P=0.002$]. On postoperative day one, the rate of opioid usage (OME/kg/h) begins to decrease in the patient-controlled analgesia group and increase in the spinal and epidural groups (Fig. 2B). All patient-controlled analgesia opioids were morphine or hydromorphone. Oxycodone and hydrocodone were the predominant oral opioids. Significantly less patients in the spinal group (18%) and epidural group (3%) received opioids by patient-controlled analgesia ($P<0.0001$). Morphine was the predominant spinal opioid, although patients also received sufentanil and hydromorphone, sometimes in combination. The P-values for daily opioid comparisons are in Table 2.

For the entire cohort, total annual mean opioid usage decreased each year and by nearly half over the 11-year period [(OME/kg/h) (95% CI): 2011: 0.043 (95% CI: 0.035-0.053) vs. 2021: 0.017 (95% CI: 0.013-0.021); $P<0.0001$] (Fig. 3). Over the same time span mean pain scores declined statistically but not clinically [(VPS) (95% CI): 2011: 3.4 (95% CI: 2.8-4.1) vs. 2021: 3.1 (95% CI: 2.5-3.7); $P<0.0001$]. The use of multimodal analgesia (yes/no) increased significantly during the same time span [2012: 118/132 (80%) vs. 2021: 173/173 (100%), $P<0.0001$].

We found evidence for small but statistically significant differences in opioid usage for all demographic

TABLE 1. Demographic Data in the 3 Treatment Groups

Group	Total	Number hospitals	Age (y) (SD)	Sex % (F/M)	BMI (SD)	Mean number implanted hardware (SD)	Race percentage (W/AA/A/O)
EPI	263	1	15.4 (1.7)	80.0/20.0	21.4 (4.0)	36.2 (5.9)	89.4/1.5/0.4/8.7
PCA	2018	11	15.3 (1.5)	74.4/25.6	22.6 (5.2)	35.2 (10.3)	70.8/12.8/1.2/15.2
SPI	90	2	16.4 (1.8)	79.6/20.4	21.5 (4.4)	39.5 (9.7)	35.6/3.3/14.4/46.7

Race: W = White, AA = African American, A = Asian, O = other.
BMI indicates body mass index; EPI, continuous epidural analgesia; PCA, patient-controlled analgesia; SPI, spinal opioid analgesia.

factors. Females used 10% (95% CI: 5%-14%) more opioids than males. The rate of opioid usage increased with age at a rate of 4% (95% CI: 3%-5%) per additional year. Opioid usage decreased with increasing body mass index 2% per unit (95% CI: 1.6%-2.4%) and increased across postoperative day. Whites used 9.4% (95% CI: 0.3%-18.7%) more opioids than African Americans.

Mean daily pain scores were < 4 for the entire 88 hours in all groups. Mean daily pain scores were statistically significantly lower, but not clinically significant,[11] on the day of surgery and postoperative day one in the spinal group when compared with the patient-controlled analgesia group (Fig. 4) [DOS: PCA 3.3 (95% CI: 2.9-3.7) vs. SPI 2.0 (95% CI: 1.3-2.7); $P = 0.001$; POD 1: PCA 3.8 (95% CI: 3.4-4.2) vs. SPI 2.9 (95% CI: 2.2-3.6); $P = 0.03$]. Pain scores trended upward each day through postoperative day 2, then declined, except in the epidural group, where mean pain scores increased after discontinuing the epidural on postoperative day 3.

There were no statistically significant differences in postoperative complications for the entire 88-hour study window in any group. The incidence of respiratory depression and excessive sedation were statistically significantly increased in the spinal group compared with the epidural group on the day of surgery ($P = 0.01$). The incidence of nausea and vomiting was statistically significantly less in the epidural group compared with the patient-controlled analgesia groups on postoperative day 1 ($P = 0.01$).

The length of stay was not statistically significantly different in the 3 groups. The epidural and patient-controlled analgesia groups went home on postoperative day 4, the spinal group went home on postoperative day 5 [(Hours): EPI 115 (95% CI: 81.7-163.8), PCA 95.7 (95% CI: 85.9-106.7), SPI 129.4 (95% CI: 100.9-166.1); $P = 0.07$]. The length of stay for the entire cohort declined by 23% over the 11-year time span [(Hours): 2011 129.5 h (95% CI: 110.2-152.2) vs. 2021 99.7 h (95% CI: 84.9-117.2), $P < 0.001$]. The documentation of first ambulation was significantly longer in the spinal opioid group compared with the patient-controlled analgesia group only [(Hours): EPI 27.9 (95% CI: 23-59) vs. SPI 74.0 (95% CI: 51.1-96.9) vs. PCA 41.7 (95% CI: 31.8-51.6); $P = 0.04$]. Time until ambulation declined by 32% over the 11-year span [2011 57.5 h (95% CI: 42.8-72.2) vs. 2021 39.1 (95% CI: 24.4-53.7); $P < 0.001$]. The time to remove the Foley catheter and intravenous line decreased by 50% to 60% during the 11-year period.

There were 32 patients (1.3%) who were readmitted to the hospital within 30 days of the surgical procedure. Postoperative wound infection requiring readmission occurred in 20 patients (0.8%); 13 of these required surgical debridement. There were 2 hardware failures requiring surgical revision. Constipation requiring rehospitalization occurred in 6 patients. One patient each was readmitted with anemia, inadequate pain control, pyelonephritis, and hypocalcemia with DeGeorge syndrome. The treatment group did not influence the readmission rate.

DISCUSSION

Opioid based analgesia has been the primary method of pain control following posterior spinal fusion since the first reported spine fusion for scoliosis correction in 1909.[12] Following the first publication describing spinal opioids for posterior spinal fusion in 1988[13] and epidural analgesia in 1996,[14] which analgesic technique reduces opioid requirements, and whether this leads to decreased pain scores, complications, and length of stay has been difficult to answer. The primary finding of this study is that spinal opioid and continuous epidural analgesia reduce opioid requirements compared with patient-controlled analgesia. However, this does not lead to a clinically significant difference in pain scores, complications or length of stay between the groups.

Beginning on postoperative day one, we observed decreasing opioid requirements in the patient-controlled analgesia group and increasing opioid requirements in the spinal and epidural groups. This may be explained by the effectiveness of opioids in animal and in vitro studies of inflammatory pain.[15] On the day of surgery, the epidural and spinal groups have reduced opioid requirements due to the somatic analgesia provided by the neuraxial block.[16] On postoperative day one inflammatory mediated pain increases as accumulating proinflammatory mediators (ie, prostaglandins and cytokines) sensitize peripheral pain sensing neurons.[17,18] The patient-controlled analgesia group achieves effective plasma level of opioids in the first 24 hours and opioid requirements begin to decline.[19] The epidural and spinal groups experience increasing opioid requirements on postoperative day one due to incomplete regional analgesia and inadequate plasma levels of opioid analgesia. The epidural and spinal groups lose somatic pain relief when the neuraxial block recedes and spend the next 24 hours attempting to reach an effective steady state of opioids to achieve adequate analgesia.

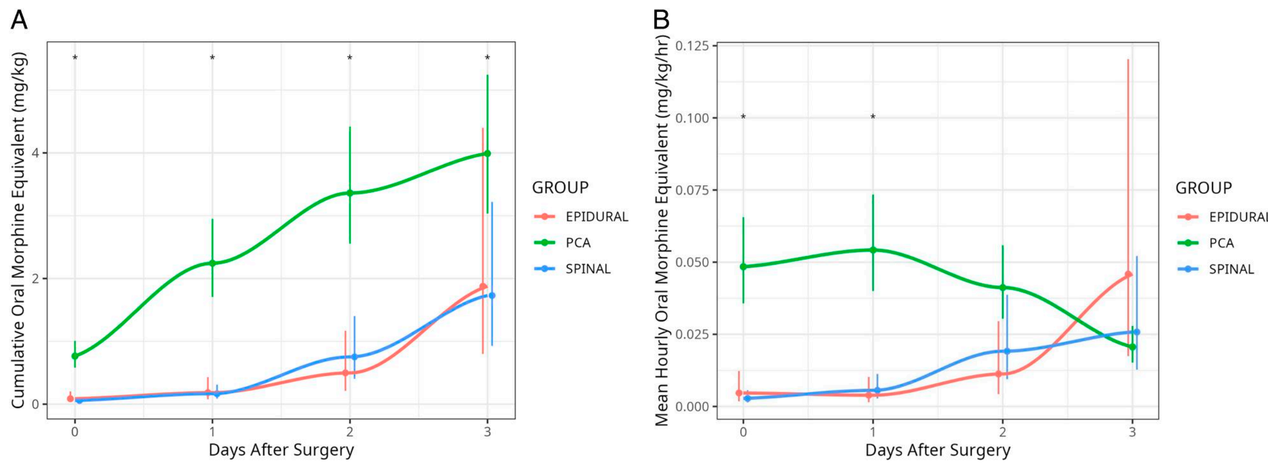

FIGURE 2. (A) Mean daily opioid usage in the 3 treatment groups. $^{*}P<0.05$. (B) The mean hourly rate of opioid usage in the 3 treatment groups. Epidural indicates continuous epidural analgesia; PCA, patient-controlled analgesia; spinal, spinal opioids. $^{*}P<0.05$. full color online

Opioid stewardship initiatives have encouraged physicians to reduce opioid usage postoperatively. Whether opioid usage declined and its effect on postoperative pain scores has been unclear. In the entire cohort, opioid usage decreased annually and by nearly half over the 11-year span, while mean pain scores were not clinically different. The evidence we found for increasing use of multimodal analgesia, earlier ambulation and enhanced recovery after surgery protocols may have contributed to maintenance of analgesia with less reliance on opioid administration.

We found statistically significant increases in the incidence of excessive sedation and respiratory depression in the spinal group compared with the epidural group on the day of surgery, and decreased nausea and vomiting in the epidural group compared with the patient-controlled analgesia group on postoperative day one. However, none of these complications were statistically significantly different over the entire 88-hour study period, caused an increase in reported serious adverse events or affected the length of stay. Recent studies have reported patients with adolescent idiopathic scoliosis receiving patient-controlled analgesia going home on postoperative day 1 or 2.[20] This length of stay appears obtainable due to decreasing opioid requirements beginning on postoperative day 1 in the opioid treatment group, when opioid requirements are still increasing in the epidural and spinal groups.

TABLE 2. Mean Opioid Usage Comparisons By Postoperative Day in the 3 Treatment Groups

POD	Treatment group	Estimated difference (SE)	P
DOS	EPI vs. PCA	−2.2 (0.45)	< 0.001*
DOS	EPI vs. SPI	0.39 (0.54)	0.74
DOS	PCA vs. SPI	2.57 (0.34)	< 0.001*
POD 1	EPI vs. PCA	−2.50 (0.45)	< 0.001*
POD 1	EPI vs. SPI	0.10 (0.54)	0.98
POD 1	PCA vs. SPI	2.60 (0.34)	< 0.001*
POD 2	EPI vs. PCA	−1.91 (0.45)	< 0.001*
POD 2	EPI vs. SPI	−0.41 (0.54)	0.72
POD 2	PCA vs. SPI	1.50 (0.34)	< 0.001*
POD 3	EPI vs. PCA	−0.75 (0.45)	0.22
POD 3	EPI vs. SPI	0.08 (0.53)	0.99
POD 3	PCA vs. SPI	−0.83 (0.34)	0.04*

*Statistical significance < 0.05.
CEA indicates continuous epidural analgesia; DOS, day of surgery; OBA, opioid based analgesia; POD, postoperative day; SBA, spinal-based analgesia.

We included only parenteral and oral opioids administered postoperatively. There is a lack of evidence-based guidelines in converting neuraxial opioids to oral morphine equivalent doses.[21] Some authors suggest oral morphine conversion ratios of 100 mg IV, 10 mg epidural and 1 mg spinal; however, these are based on highly opioid tolerant adult patients converting from oral to the neuraxial route. Should we apply the same ratio in reverse when converting from neuraxial to parenteral and oral opioids in opioid naive pediatric patients? We chose to focus on parenteral and oral opioids with well-established conversion to oral morphine equivalents and which are responsible for the most concerning side-effects of respiratory depression, excessive sedation and decreased gastrointestinal motility.

We were forced to remove 5 facilities due to inaccuracies in data entry. Studies involving large data sets illustrate the importance of accurate data. Health care institutions should initiate quality improvement initiatives to monitor and improve the accuracy of data so that future studies can provide meaningful findings.

The increased time to ambulation in the spinal opioid group was a surprising finding. Previous studies of spinal opioids for spine surgery in adults and children have reported earlier time to ambulation, decreased opioid requirements and earlier hospital discharge.[22] This finding may reflect the increased sedation observed in the spinal group on the day of surgery, a delayed ambulation

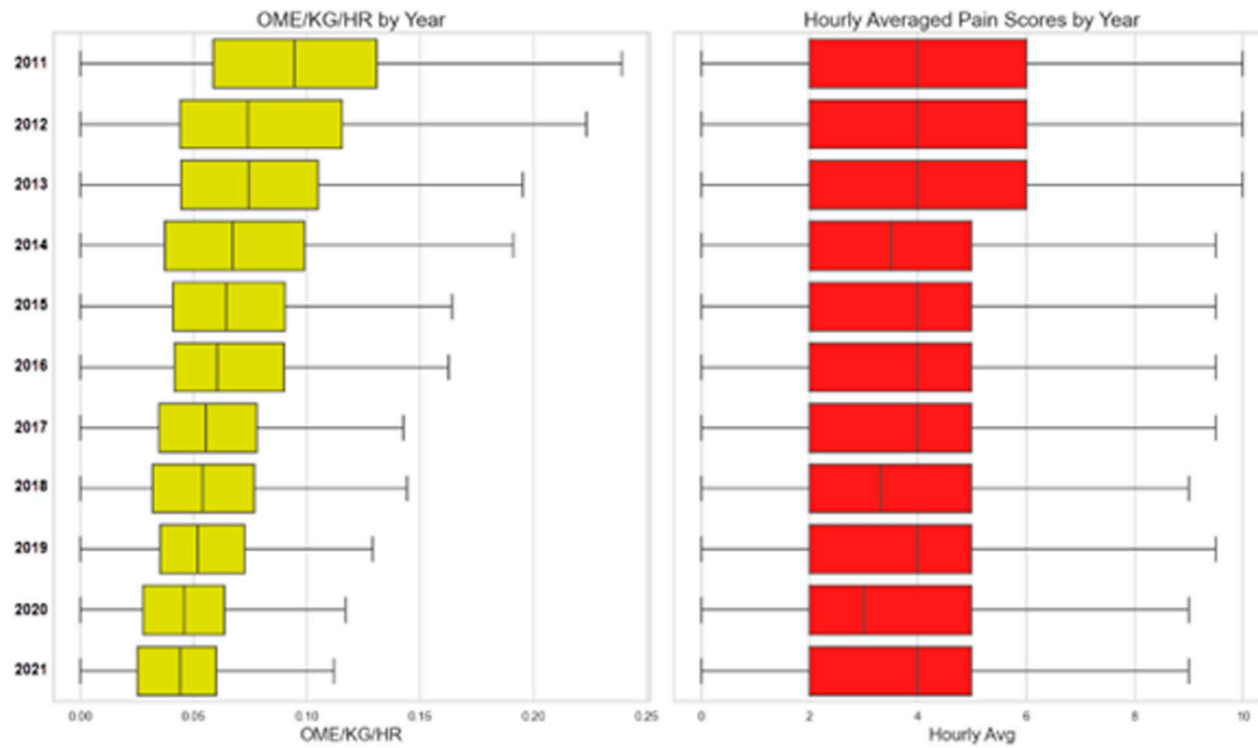

FIGURE 3. Annual mean opioid usage and pain scores for the entire cohort (years: 2011 to 2021). OME/kg/hr indicates oral morphine equivalents/kg/hour.

protocol specific to the 2 spinal opioid locations or reflect differences in nursing practice in documentation of ambulation. The length of stay was not significantly affected by the delay in ambulation.

This study has limitations typical of multicenter cohort studies. The considerable variability at the facility-level presents challenges in distinguishing between the effects of the facility and the overall effects of the method of analgesia, especially with only a single hospital represented in the epidural group. Hospital-level variability of this magnitude suggests that analgesic protocols at individual hospitals are important in determining opioid usage following posterior spinal fusion, independent of the method of analgesia. We used multilevel regression analysis to adjust for facility and demographic differences to confirm that epidural and spinal-based analgesia decrease opioid requirements in the postoperative period. Beginning on postoperative day one, epidural and spinal analgesia show waning effectiveness, most likely secondary to an inability to respond to an increase in inflammatory pain. We conclude that all 3 methods of analgesia provide safe and effective pain relief. This study reinforces the need for collaboration between pediatric orthopaedic surgeons and anesthesiologists to effectively manage postoperative pain following posterior spinal fusion for adolescent idiopathic scoliosis.

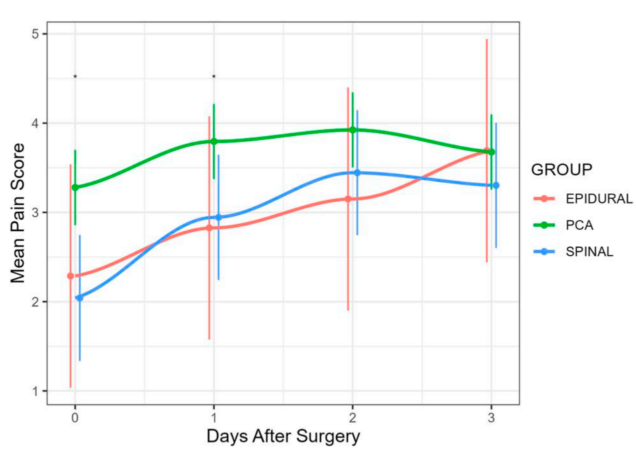

FIGURE 4. Mean daily pain scores in the 3 treatment groups. Epidural indicates continuous epidural analgesia; PCA, patient-controlled analgesia; Spinal, spinal opioids. *$P<0.05$.

After this study's completion, a patient at the epidural site developed permanent paralysis after her epidural catheter was dosed postoperatively with a bolus of local anesthetic in the postanesthesia care unit. She was diagnosed with epidural lipomatosis and a spinal cord infarct from mechanical compression of the spinal cord from the combined effect of epidural fat and local anesthetic volume. She underwent a decompressive laminectomy but has not regained any neurological function below the eighth thoracic vertebral level. It is a reminder that while continuous epidural analgesia is exceedingly safe, the risk is not zero. No Shrine hospital locations are currently using epidural analgesia for posterior spinal fusion.

ACKNOWLEDGMENTS

The authors thank Charlotte Preuschoff, Clinical Research Coordinator, Research Department, Shriners Children's Hospital, Spokane, Washington, for her assistance in data collection and coordination for this study.

REFERENCES

1. Blumenthal S, Min K, Nadig M, et al. Double epidural catheter with ropivacaine versus intravenous morphine: a comparison for postoperative analgesia after scoliosis correction surgery. *Anesthesiology*. 2005;102:175–180.
2. Sucato DJ, Duey-Holtz A, Elerson E, et al. Postoperative analgesia following surgical correction for adolescent idiopathic scoliosis: a comparison of continuous epidural analgesia and patient-controlled analgesia. *Spine*. 2005;30:211–217.
3. Hong R, Gibbons K, Li Y, et al. A retrospective comparison of intrathecal morphine and epidural hydromorphone for analgesia following posterior spinal fusion in adolescents with idiopathic scoliosis. *Ped Anesth*. 2016;27:91–97.
4. Ibach BW, Loeber C, Shukry M, et al. Duration of intrathecal morphine effect in children with idiopathic scoliosis undergoing posterior spinal fusion. *J Opioid Manag*. 2015;11:295–303.
5. Klatt JW, Mickelson J, Hung M, et al. A randomized prospective evaluation of 3 techniques of postoperative pain management after posterior spinal instrumentation and fusion. *Spine*. 2013;38:1626–1631.
6. Cassady JF, Lederhaas G, Cancel DD, et al. A randomized comparison of the effects of continuous thoracic epidural analgesia and intravenous patient-controlled analgesia after posterior spinal fusion. *Reg Anesth*. 2000;25:246–253.
7. Van Boerum, Smith JT, Curtin MD. A comparison of the effects of patient-controlled analgesia with intravenous opioids versus epidural analgesia on recovery after surgery for idiopathic scoliosis. *Spine*. 2000;25:2355–2357.
8. Nielsen S, Degenhardt L, Hoban G, et al. A synthesis of oral morphine equivalents for opioid utilization studies. *Pharmacoepidemiol Drug Saf*. 2016;25:733–737.
9. R Core Team. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing; 2023https://www.R-project.org/
10. Lenth R (2023). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package version 1.8.8, <https://CRAN.R-project.org/package = emmeans
11. Issa TZ, Lee Y, Henry TW, et al. Values derived from patient reported outcomes in spine surgery: a systematic review of the minimally important clinical difference, substantial clinical benefit, and patient acceptable symptom state. *Eur Spine J*. 2023;32: 3333–3351.
12. Tarpada SP, Morris MT, Burton DA. Spine fusion surgery: a historical perspective. *J Orthop*. 2017;14:134–136.
13. Dalens B, Tanguy A. Intrathecal morphine for spinal fusion in children. *Spine*. 1988;13:494–498.
14. Shae BA, Watson TI, et al. The safety of continuous epidural infusion for postoperative analgesia in pediatric spine surgery. *J Pediatr Orthop*. 1996;16:374–377.
15. Stein C, Kuchler S. Non-analgesic effects of opioids: peripheral opioid effects on inflammation and wound healing. *Curr Pharm Des*. 2012;18:6053–6069.
16. Slappendel MJ, Gielen MJ, Hasenbos MA, et al. Fm. Spread of radiopaque dye in the thoracic epidural space. *Anaesthesia*. 1988;43: 939–942.
17. Kidd BL, Urban LA. Mechanisms of inflammatory pain. *Br J Anaesth*. 2001;87:3–11.
18. Lipnik-Strangelj M. Mediators of inflammation as targets for chronic pain treatment. *Mediators Inflamm*. 2013;2013: 78235.
19. Reder RF, Oshlack B, Miotto JB, et al. Steady-state bioavailabiity of controlled-release oxycodone in normal subjects. *Clin Thera*. 1996;18:95–105.
20. Gorlin AW, Rosenfield DM, Maloney J, et al. Survey of pain specialists regarding conversion of high-dose intravenous to neuraxial opioids. *J Pan Res*. 2016;21:692–700.
21. Glass PS, Estock P, Ginsberg P, et al. Use of patient-controlled analgesia to compare the efficacy of epidural to intravenous fentanyl administration. *Anesth Analg*. 1992;74:345–351.
22. Sarwahin V, Hasan S, Liao B, et al. Zero patient-controlled analgesia is an achievable target for postoperative rapid recovery management of adolescent idiopathic scoliosis patients. *Spine*. 2021; 46:1448–1454.